



Unit 4 · Outcome 2 · Energy Sources

Non-renewable energy sources

KK01

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



DEFINITION

What “non-renewable” actually means

A non-renewable energy source is one that exists in a finite stock and is replenished by natural processes far more slowly than humans consume it — effectively not at all on a human timescale. Once you burn this year’s coal, the Earth will not make more for tens of millions of years.

The contrast is with renewable sources (sun, wind, flowing water, tides, geothermal heat, biomass), which are replenished on a human timescale — the sun rises again tomorrow. The dividing line isn’t “will it run out one day” (the sun will too, in ~5 billion years); it’s the rate of natural replenishment versus the rate of human use.

◆ ANALOGY

🏠 Analogy — inheritance vs salary Treat the Earth’s non-renewables like a one-off inheritance sitting in a savings account: a huge balance, but no deposits are coming. Every dollar you spend lowers the total forever. Renewables are a salary — money that lands in the account every fortnight whether you spend it or not. A society running on non-renewables is living off savings; a society running on renewables is living within its income. The whole Energy outcome is really about shifting Australia from the first to the second.

Coal · oil · natural gas · coal seam gas · nuclear (uranium). Formed over millions of years; cannot be regrown within human time. Using them depletes the resource.

⚠ WATCH OUT

The trap hiding in this dot point Four of the five sources here are fossil fuels (coal, oil, natural gas, coal seam gas) — but the fifth, nuclear, is non-renewable yet NOT a fossil fuel. “Non-renewable” and “fossil fuel” are not synonyms. Keep that distinction primed; it’s tested almost every year.

D · SCRUB THE TIMELINE

Deep time: how fossil fuels formed

The four fossil fuels are, quite literally, ancient sunshine. Hundreds of millions of years ago, photosynthesis locked solar energy into the chemical bonds of living things. When that organic matter was buried before it could fully decay, then cooked under heat and pressure for geological ages, the carbon-rich remains became coal, oil and gas. Burning them today releases that stored sunlight — and the stored carbon.

Starts as dead vegetation in ancient swamps → compressed to peat → lignite (brown coal) → bituminous (black coal) → anthracite. More heat, time and pressure = higher carbon content and energy density. Victoria’s Latrobe Valley sits on enormous brown coal seams (younger, wetter, lower energy density).

Marine micro-organisms (plankton, algae) settle on ancient sea floors → buried in sediment → become a waxy substance called kerogen → moderate heat yields oil; greater heat “cracks” it into natural gas. The hydrocarbons migrate upward until trapped under impermeable rock — that’s a reservoir.

◆ KEY INSIGHT

✓ Carry-forward fact Because fossil fuels store carbon that was removed from the atmosphere millions of years ago, combusting them returns long-buried CO₂ to today’s atmosphere. That single sentence is the bridge to KK03 (fossil fuels & the carbon cycle) and the whole Climate chapter.

DRAG TO SORT

The line students blur

◇ ACTIVE RECALL

Two confusions sink marks every year: (1) treating “non-renewable” and “fossil fuel” as the same thing (nuclear breaks that), and (2) thinking coal seam gas is a renewable or a totally different fuel (it’s the same methane as natural gas, just unconventional). Sort these chips into the right bin to lock it in.

◆ KEY INSIGHT

✓ One-line distinctions to memorise Non-renewable $\supset$ fossil fuels + nuclear. • CSG vs natural gas: same gas (methane), different source — natural gas pools in a reservoir; CSG is held in coal seams and needs dewatering to release. • Nuclear isn’t renewable just because it’s low-carbon — uranium is a finite mined resource.

D · WATCH THE CHAIN

From fuel to the power point

Here’s the thread that connects four of the five: coal, gas, CSG and even nuclear ultimately just make heat to boil water and spin a turbine. The fuel differs; the back end is almost identical. That’s why this dot point sets up the energy-conversion and efficiency work in KK05–KK06.

◆ ANALOGY

🔗 Analogy — every thermal station is a kettle + pinwheel Picture a giant kettle (the boiler) with a pinwheel (the turbine) held in its steam. Whatever heats the kettle — burning coal, burning gas, or splitting uranium — the job is the same: make steam, spin the wheel, turn the generator. A gas turbine is the exception: hot combustion gases push the blades directly, no kettle required, which is why gas plants start so fast.

COMPARE TABLE

Advantages & disadvantages — the exam gold

VCAA almost always asks you to compare sources or justify a choice against criteria. Learn this grid as a mental scaffold: pick two or three rows that matter for the question and contrast them. Don’t list everything — select the relevant criteria.

Criterion	Coal	Oil	Natural gas	Coal seam gas	Nuclear
Renewable?	No	No	No	No	No
Fossil fuel?	Yes	Yes	Yes	Yes	No
Main use	Electricity	Transport	Electricity, heat, LNG	Electricity, LNG	Electricity
Energy density	Moderate	High	Moderate	Moderate	Very high
Operating CO ₂	Highest	High	~½ of coal	~½ of coal*	Near-zero
Dispatchable?	Yes (baseload)	Yes	Yes (fast peaker)	Yes	Yes (baseload)
Key waste/ impact	Ash, mine, air pollution	Spills, emissions	Fugitive methane	Groundwater, salt, fracking	Radioactive waste
Status in Australia	Declining	Mostly imported	Major; transitional	Major (QLD)	Prohibited

* CSG's combustion emissions match natural gas; its overall footprint depends heavily on fugitive-methane leakage during extraction.

REAL FIGURES

The Australian & Victorian picture (2025–26)

Examiners reward students who anchor general claims in real data. Here are current, defensible numbers — use them, but quote the trend as much as the exact figure.

of Australia's 2025 electricity from fossil fuels — coal 42.7%, gas 16.2%, oil 1.7%. Renewables made up the other 39.5%.

of Victoria's electricity still comes from brown coal (Latrobe Valley) — the most CO₂-intensive fuel, now being phased out.

nuclear generation — prohibited under the EPBC Act 1999 & ARPANS Act 1998, despite Australia exporting uranium.

Three brown-coal stations remain, all with announced closure dates: Yallourn (EnergyAustralia) closing mid-2028, Loy Yang A (AGL) by 2035, and Loy Yang B (Alinta) intending to run until ~ 2047. Each closure is paired with new storage (e.g. big batteries) to firm renewables — a concrete example of the energy transition this whole outcome examines.

Coal: brown in VIC, black in NSW/QLD. Oil & conventional gas: Bass Strait, Cooper Basin (Moomba), North West Shelf — declining domestically. Coal seam gas: mainly Queensland's Surat/Bowen basins feeding Gladstone LNG, with reserves (and moratoria debates) in NSW & VIC. Nuclear: uranium mined in SA & NT for export only.

EXAM CRAFT

Decode the command word first

◆ COMMAND WORDS

Before writing a word, identify what the verb is asking for. The same content earns very different marks depending on whether you were told to describe or to evaluate. State it — no explanation. “Name two non-renewable energy sources.” → Coal and natural gas. Give the features/characteristics. What it is and how it works — no judgement. State the difference between two things, ideally on the same criterion (e.g. source of CSG vs natural gas). Give both similarities and differences, point-against-point — not two separate paragraphs. Give reasons / a causal chain. “Why does gas emit less CO₂ than coal?” Weigh advantages against disadvantages against stated criteria, then reach a supported judgement.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

MODEL ANSWER BUILDER

Writing like a top student — the P·E·L move



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

🔗 P·E·L WRITING

For any “explain / compare / justify” item, structure each idea as P·E·L — Point, Evidence, Link. Make the claim, back it with a specific fact or figure, then link it explicitly to the question. Walk the tabs below to see a 3-mark answer assembled.

AVOID THESE

Six traps that quietly cost marks

⚠ EXAM TRAPS

✗ “Nuclear is a fossil fuel.” ✓ Nuclear is non-renewable but NOT a fossil fuel — it uses fission of mined uranium, not combustion of buried organic matter. ✗ “Coal is renewable because we can keep mining it.” ✓ Renewability is about the rate of natural replenishment, not whether stock remains. Coal reforms over millions of years — far slower than we use it. ✗ “CSG is a different fuel from natural gas.” ✓ Same fuel (methane) — only the source differs: a reservoir (natural gas) vs adsorbed in coal seams (CSG). ✗ “Natural gas is clean / green / renewable.” ✓ It’s a lower-emission fossil fuel, still finite and still emitting CO₂ (plus methane if it leaks). “Cleaner than coal” ≠ “clean”. ✗ “Nuclear has no environmental downside because it emits no CO₂.” ✓ Its key disadvantage is long-lived radioactive waste (plus cost, build time and uranium being finite). ✗ Listing every advantage you know. ✓ Select criteria relevant to the question and compare point-against-point. A wall of facts with no comparison scores poorly on “compare/evaluate”.

SELF-MARK · 24 MARKS

Practice questions

★ PRACTICE & SELF-MARK

VCAA-style items escalating from 2 to 6 marks. Attempt each on paper first, then reveal a weak response, an annotated strong response, and the marking guide. Award yourself marks honestly — the dock bottom-right tracks your total. “Runs out” misses the rate idea; solar is renewable — a careless naming error costs a mark, and they’ve omitted coal seam gas.

- 1 — definition conveys finite AND replenished slower than used / not on a human timescale.
- 2 — all five correct (oil, coal, natural gas, coal seam gas, nuclear). 1 mark if 3–4 correct; 0 if a renewable is included as a main error.

★ PRACTICE & SELF-MARK

Circular and vague. No mention of adsorption/reservoirs or dewatering; “pollution” isn’t specific to CSG.

- 1 — natural gas located in a reservoir / free gas.
- 1 — CSG adsorbed in coal seams / released by dewatering.
- 1 — valid CSG advantage. 1 — valid CSG-specific disadvantage (groundwater, saline water, fracking, fugitive methane).

★ PRACTICE & SELF-MARK

The CO₂ point isn’t the reason it’s non-renewable; part b states a fact but never explains the two classifications.

- 1 — identifies the “renewable” claim as the error. 1 — correct reasoning (finite uranium / replenishment rate).
- 1 — non-renewable justified (finite uranium). 1 — not-fossil justified (fission, not combustion of organic matter).

★ PRACTICE & SELF-MARK

No named energy forms, no turbine/generator step; part b lacks a specific consequence.

- a: 1 — heat boils water to steam; 1 — steam (kinetic) turns turbine; 1 — generator produces electrical energy. (Naming energy forms required for full marks.)
- b: 1 — combustion vs fission as the heat source; 1 — a valid matching consequence (CO₂/pollution vs radioactive waste).

★ PRACTICE & SELF-MARK

Right instinct, but it's a thin list. No figures, no point-against-point comparison, criteria not clearly applied, judgement asserted not justified. Maybe 3/8. Emissions: Both are non-renewable, but gas emits roughly half the CO₂ per unit electricity of the closing brown coal, while nuclear produces near-zero operational CO₂. On emissions alone, nuclear is superior. Build time & transition fit: A gas peaker can be built and dispatched in a few years and starts within minutes, ideal for firming variable renewables now; a nuclear plant takes well over a decade, so it could not provide dispatchable supply during this transition. Waste & cost: Gas avoids long-lived waste but locks in ongoing CO₂ and fugitive-methane risk; nuclear avoids CO₂ but creates long-lived radioactive waste at high cost. Legality: Decisively, nuclear is prohibited in Australia (EPBC Act 1999; ARPANS Act 1998), so it is not currently a real option. Judgement: For reliability during the transition, the gas peaker is the better non-renewable option: it is fast to deploy, dispatchable and legal, and roughly halves emissions versus coal — even though nuclear scores better on operational emissions, its decade-plus build time and current prohibition make it unable to meet the transitional need.

- 1–2 — both correctly identified as non-renewable; relevant criteria selected.
- 3–5 — genuine point-against-point comparison on ≥3 criteria with specific evidence (CO₂ ~½ coal, build times, waste type, legality).
- 6–7 — emissions vs reliability tension handled; transition context applied.
- 8 — clear, supported judgement that follows from the comparison.

REVISE FAST

One-screen cheat sheet

- Finite stock; replenished far slower than used → not on a human timescale.
- Renewable = replenished in-time (sun, wind, water, geothermal, biomass).
- Oil · Coal · Natural gas · Coal seam gas · Nuclear.
- First four = fossil fuels; nuclear = non-fossil non-renewable.
- Coal = buried plants (peat→lignite→bituminous→anthracite).
- Oil & gas = buried plankton → kerogen → oil (then gas with more heat).
- CSG = methane adsorbed in coal seams (dewater to release).
- Nuclear = fission of finite mined uranium.
- Coal & nuclear: heat → steam → turbine → generator.
- Gas: gas turbine (fast peaker); oil: mostly transport fuel.
- Coal: cheap, reliable / highest CO₂.

- Gas: $\sim\frac{1}{2}$ coal's CO_2 , dispatchable / still fossil, methane leaks.
- Nuclear: near-zero CO_2 / radioactive waste, slow, banned in AUS.
- Fossil 60.5% (coal 42.7, gas 16.2, oil 1.7); renewables 39.5%.
- VIC $\sim$ 56% brown coal; Yallourn 2028, Loy Yang A 2035.
- Nuclear prohibited (EPBC 1999; ARPANS 1998).

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

point Gas is better than coal. (Vague — “better” how? No criterion.)

✓ STRONG / MODEL ANSWER

evidence Burning methane releases roughly half the CO_2 per unit of electricity that brown coal does, and gas turbines can start within minutes, so they can firm variable solar and wind when output drops.

✗ WEAK ANSWER

evidence It's cleaner and pretty quick. (No figure, no mechanism — examiner can't award the mark.)

✓ STRONG / MODEL ANSWER

link This lets gas bridge the period between a coal-dominated grid and a fully renewable one — reducing emissions now without sacrificing reliability — which is why it's labelled “transitional” rather than a long-term solution.

✗ WEAK ANSWER

link So that's why people use gas. (Doesn't reconnect to “transitional”.)

✓ STRONG / MODEL ANSWER

Full P-E-L response Natural gas is called a “transitional” fossil fuel because it emits far less CO_2 than coal yet remains dispatchable. Burning methane releases roughly half the CO_2 per unit of electricity of brown coal, and fast-starting gas turbines can firm variable solar and wind within minutes. This lets gas bridge the shift from a coal-heavy grid to a renewable one — cutting emissions now without risking reliability — so it is treated as a stepping stone rather than a permanent, sustainable solution.